CLOUD-INTEGRATED IOT AND ML-BASED WASTE MANAGEMENT SYSTEM FOR RAILWAY STATIONS

25-26J-191

Project Proposal Report

Thisara Nuwanthi

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Nuwanthi B.D.T.	IT22566102	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: Date:

Abstract

This research details a comprehensive, cloud-integrated, IoT-based autonomous robotic system designed to address the significant waste management challenges at Sri Lankan railway stations. The current manual waste disposal methods are inefficient and fail to keep up with real-time rubbish accumulation, often leading to unhygienic conditions and overflowing bins, especially during rush hours and holidays. The lack of technological adaptation in these environments is a major shortfall that this project aims to rectify.

The proposed solution integrates robotics, IoT, and machine learning to automate the entire waste management lifecycle, from collection to segregation, with minimal human intervention. A key focus of this project is the Garbage Classification Model, which addresses the critical task of automated waste sorting. This component involves developing a machine learning model to classify waste into four predefined categories: Plastic, Paper, Glass, and Metal. My work centers on this model, which uses a custom-trained Convolutional Neural Network (CNN) for high-accuracy classification. The process involves preparing a custom image dataset of waste items from railway environments, followed by model training and optimization into a lightweight format for efficient deployment on edge devices like the ESP32-CAM. This ensures real-time operation at the disposal platform. The model's output directly controls a stationary robotic arm, which sorts the waste into the correct bins. This component aligns with the Information Technology (IT) specialization by applying machine learning, artificial intelligence, and embedded systems to create a responsive and intelligent solution. It is a critical part of the larger system, demonstrating how intelligent automation can significantly improve efficiency of sanitation, reduce manual labor, and support the country's sustainability and smart city goals.

Keywords: Waste Classification, Machine Learning, Convolutional Neural Networks, Smart Waste Management, Embedded Systems, IoT, Robotics, Real-time Sorting, Railway Stations, Sustainability.

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• The system shall preprocess the captured images (resizing, normalization, noise reduction). • The system shall classify waste into four categories: plastic, paper, glass, and

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
AI	Artificial Intelligence
SLAM	Simultaneous Localization and Mapping
IR Sensor	Infrared Sensor
GPS	Global Positioning System IMU
Inertial Measurement Unit	
QA	Quality Assurance
IDE	Integrated Development Environment
MS Planner	Microsoft Planner

1. INTRODUCTION

1.1 Background & Literature Review

Automated and intelligent waste management is increasingly critical for public transit locations such as railway stations, which face high and fluctuating passenger volumes and diverse waste streams [1]. In Sri Lanka, stations still largely depend on manual collection that is unable to dynamically respond to real-time waste accumulation, leading to overflowing bins, poor hygiene, and negative public perceptions [2]. Numerous studies have demonstrated that integrating deep learning and IoT technologies into waste management systems can bring substantial improvements to operational efficiency, resource recovery, and sustainability [3][4].

Deep learning methods, especially convolutional neural networks (CNNs), have shown remarkable potential for automatic, real-time classification of waste categories.Plastic, paper, glass, and metal even in complex settings like railway stations or city streets [1]. A recent study by Alourani et al. (2025) demonstrated that smart bins equipped with cameras and deep neural networks achieved up to 99% classification accuracy for solid waste categories, far surpassing conventional rule-based or sensor-only approaches. Similarly, Le et al. (2025) showed that coupling robotic systems with deep learning waste classifiers resulted in precise and efficient waste segregation in public environments, enabling end-to-end automation.

Deploying IoT and cloud-based analytics further enhances the waste management lifecycle by enabling fill-level monitoring, optimized collection routes, and live system feedback (PLOS ONE, 2024). These advancements are transforming the traditional manual systems and aligning public infrastructure with global “smart city” and sustainability frameworks (Kannan et al., 2024).

Harnessing state-of-the-art AI in railway waste classification not only improves recycling rates and reduces labor costs but also contributes to a cleaner, more sustainable urban environment, fully justifying the development of an advanced machine learning waste classification solution.

1.2 Research Gap

Waste management in Sri Lanka continues to face significant challenges due to the absence of advanced technological solutions for collection and classification. Although various reforms have been introduced, studies reveal that the country's municipal waste systems remain heavily dependent on manual labor with very little automation. Gunasekara and Silva (2021) note that these systems lack efficiency, while Bandara et al. (2021) emphasize that poor segregation practices limit recycling opportunities and worsen landfill dependency. This situation is particularly problematic in high-traffic areas such as railway stations, where the constant accumulation of plastics, paper, food packaging, and glass demands timely and systematic classification. However, no automated classification mechanisms are currently in place to address these pressing issues [1][2].

Globally, rapid progress has been made in applying smart technologies to waste management. Ahmed et al. (2022) highlight how IoT and machine learning systems have been deployed in smart cities to improve waste monitoring and collection. Similarly, Hannan et al. (2021) introduced IoT-enabled smart bins that transmit fill-level data in real time, thereby enabling better scheduling of waste collection. While such solutions demonstrate clear improvements in urban waste handling, they are often designed for structured or controlled environments. This limits their applicability to crowded and dynamic spaces such as Sri Lankan railway stations, where unpredictable littering and fluctuating pedestrian volumes make the problem far more complex [3][4].

Machine learning, particularly deep learning, has emerged as a powerful tool for automatic waste classification. Rad et al. (2022) conducted a comprehensive review and confirmed that convolutional neural networks (CNNs) achieve high accuracy in distinguishing recyclables. More recently, Nahidaman et al. (2025) developed an automated waste classification system using CNNs, reporting strong results on benchmark datasets. However, a key limitation of these studies is that most rely on publicly available or simulated datasets that do not reflect the unstructured and cluttered nature of waste found in real-world railway environments. Additionally, many of these models are computationally intensive and cannot be deployed efficiently on low-power edge devices used in mobile robots or smart disposal units [5][6].

Recent advancements have begun to address these challenges. Qiu et al. (2025) introduced an enhanced EfficientNetV2 model that integrates CE-Attention and depthwise separable convolutions, achieving 95.4% accuracy with reduced complexity. Similarly, Kaur et al. (2024) demonstrated the robustness of EfficientNetV2B1 for waste classification on both balanced and unbalanced datasets, achieving accuracy rates above 95%. Meanwhile, the RealWaste dataset (2023) provides landfill images that represent messy, real-world conditions, with Inception V3 achieving 89.19% accuracy compared to only 49.69% on cleaner samples. These developments highlight the growing interest in lightweight, efficient, and realistic waste classification approaches [7][8][9].

Despite these advancements, significant gaps remain in research related to Sri Lanka's railway stations. There is no localized dataset that captures the unique types of waste generated in these environments. Moreover, very few studies focus on lightweight CNN architectures optimized for edge deployment, which is crucial for integration into robotic systems. Finally, there is limited research addressing the robustness of models in dynamic real-world conditions, where lighting, occlusion, and heavy pedestrian movement affect accuracy. This study seeks to bridge these gaps by creating a railway-specific waste dataset, developing a lightweight CNN model for classification, and validating its performance under the real environmental conditions of Sri Lankan railway stations.

Aspect Existing Systems / Research Findings Identified Research Gap > Thea: Waste Management in Sri Lanka Manual collection, scheduled cleaning, and mixed waste disposal with little to no classification [1][2]. No automation or AI-driven waste segregation in railway stations; heavy reliance on manual labor.

Smart City Waste Solutions IoT-based smart bins and fill-level monitoring applied in some urban contexts [3][4]. Mostly designed for structured environments; not tested in chaotic, high-traffic railway platforms.

Machine Learning for Waste Classification CNNs achieve strong performance (80–95% accuracy) on controlled datasets [5][6]. Models often rely on generic datasets (Kaggle, Roboflow) that do not reflect Sri Lankan waste characteristics.

Datasets Used Publicly available datasets (TrashNet, Kaggle, Roboflow) with clean and uniform images [5][6]. Lack of localized, real-world railway waste datasets with diverse lighting, clutter, and environmental conditions.

Model Complexity Deep CNNs and standard architectures (ResNet, Inception, VGG) used in prior studies [5][6]. Many models are computationally heavy, unsuitable for deployment on edge devices in mobile robots.

Recent Advances Lightweight models (EfficientNetV2, MobileNet) and RealWaste dataset show efficiency and realism [7][8][9]. Still not applied to Sri Lankan context; railway station scenarios remain untested.

Integration into Robotics Few studies integrate classification models with robotic arms for sorting; mostly proof-of-concept [3][4]. No end-to-end integration with autonomous garbage collection robots in railway station environments.

Table 1.1 Comparison of previous researches

Aspect	Existing Systems	Research Gaps Identified
Waste Management in Sri Lanka	Manual collection, scheduled cleaning, and mixed waste disposal with little to no classification [1][2].	No automation or AI-driven waste segregation in railway stations; heavy reliance on manual labor.
Smart City Waste Solutions	IoT-based smart bins and fill-level monitoring applied in some urban contexts [3][4].	Mostly designed for structured environments; not tested in chaotic, high-traffic railway platforms.
Machine Learning for Waste Classification	CNNs achieve strong performance (80–95% accuracy) on controlled datasets [5][6].	Models often rely on generic datasets (Kaggle, Roboflow) that do not reflect Sri Lankan waste characteristics.
Datasets Used	Publicly available datasets (TrashNet, Kaggle, Roboflow) with clean and uniform images [5][6].	Lack of localized, real-world railway waste datasets with diverse lighting, clutter, and environmental conditions.
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Integration into Robotics	Few studies integrate classification models with robotic arms for sorting; mostly proof-of-concept [3][4].	No end-to-end integration with autonomous garbage collection robots in railway station environments.

1.3 Research Problem

Waste management in Sri Lanka's railway stations is still handled in a largely manual manner, where garbage is collected in bulk without proper classification or segregation. This results in recyclable items such as plastic, paper, glass, and metal being discarded together, reducing recycling opportunities and contributing to environmental pollution.

Although smart waste systems and machine learning–based classification models have been developed in other contexts, they have not been adapted for Sri Lankan railway environments. The unique challenges of railway stations. Such as high passenger density, irregular littering patterns, and varying environmental conditions make existing solutions unsuitable for direct application.

Furthermore, most current classification models are trained on generic datasets and require high computational resources, which are impractical for real-time use on low-power robotic systems. Without a lightweight and context-specific classification solution, waste segregation in railway stations will continue to rely on inefficient manual labor, limiting progress towards sustainable and smart waste management.

The research problem therefore lies in the absence of an efficient, real-time, and railway-specific waste classification model that can be integrated with automated waste collection systems to support cleaner stations, improve recycling practices, and enhance commuter and tourist experiences.

2. OBJECTIVE

2.1 Main Objectives

The main objective is to develop a cloud-integrated, IoT- and machine-learning-based autonomous waste management system that can detect, collect, classify, and intelligently unload garbage in railway stations, while ensuring real-time monitoring of disposal bins and timely notifications for efficient waste handling.

2.2 Specific Objectives

- Autonomous Garbage Detection and Collection
  - To design and implement a vision-based detection mechanism supported by
  - sensors and a robotic arm for identifying and collecting garbage items in real time.
- Intelligent Navigation and Unloading Control
  - To enable the mobile robot to autonomously navigate railway station environments with obstacle and fall prevention, continuously monitor internal bin fill levels, and perform automated unloading at predefined disposal areas.
- Waste Classification Using Machine Learning
  - To develop and train a custom machine learning model capable of classifying collected waste into four categories Plastic, Paper, Glass, and Polythene and deploy it efficiently at the disposal station.
- Automated Waste Sorting and Disposal Bin Monitoring
  - To integrate a robotic arm for placing classified waste into static bins and implement IoT-based monitoring of disposal bin fill levels, with cloud-based visualization and real-time notifications for cleaning staff.

3. METHODOLOGY

3.1 System Architecture

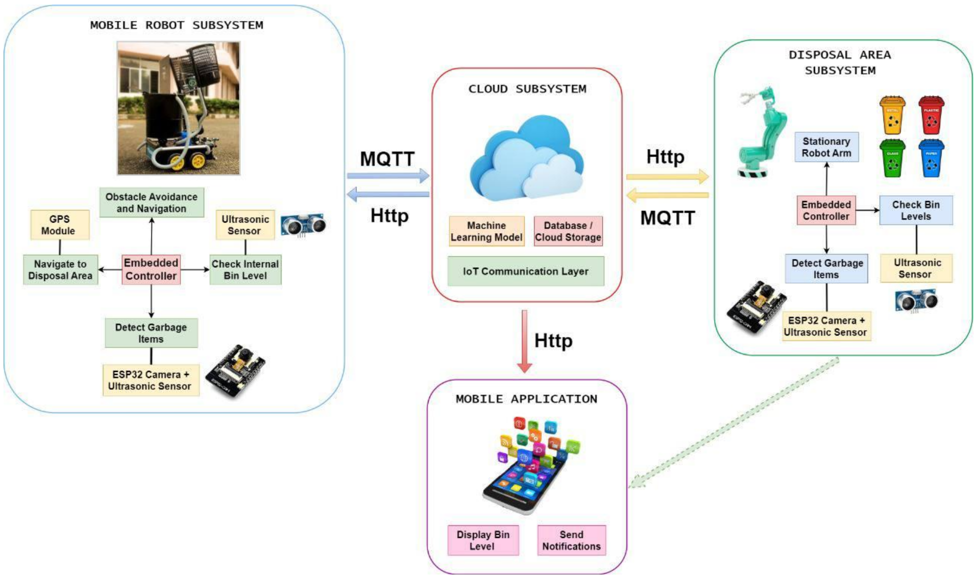

Figure 3.1: System architectural diagram for the overall system

The system has been conceptualized as an integrated waste management system which utilizes robotics, IoT, and machine learning to make the railway stations cleaner. It has been segmented into four standalone subsystems that work in flawless synergy to create a fully autonomous, efficient, and reliable system for waste collection, segregation, and disposal. Each of the subsystems offers an individual set of capabilities, but how they interoperate is what results in the achievement of the overall research objectives.

- **Mobile Robot Subsystem**: This unit offers the essence of the solution by serving as the active unit for performing the process of waste collection. It has an onboard vision system supplemented by sensors that help identify garbage objects in real-time. Upon detection of an object, the robotic arm switches on automatically to pick up the trash and deposit it into the robot's internal storage compartment. The navigation module allows the robot to move freely through station environments without facing hazards and obstacles. Safety is ensured with the use of sensors placed at strategic positions to not only navigate objects around in its path, but also to avoid falling off elevated railway platforms. The subsystem is also equipped with a vigilance system that constantly keeps an eye on the fill level of its internal container to unload only when necessary. After filling, the robot moves automatically to the disposal point with the help of location awareness and GPS.

- **Cloud Subsystem**: The cloud subsystem provides central intelligence and connectivity. It is the site of the machine learning algorithm that translates picture data to waste classification to deliver accuracy in the discrimination between types such as plastic, glass, paper, and polythene. Apart from classification, the cloud gives the communications infrastructure, connecting the mobile robot, disposal site, and mobile application into a unified architecture. It also addresses data storage, with the ability to conduct longterm trend analysis of waste and scalability to scale the solution to numerous railway stations.

- **Disposal Area Subsystem**: In the disposal area subsystem, waste handling changes from collection to segregation. Once the mobile robot off-loads the internal bin onto the platform, it is passed over by a fixed robotic arm with inbuilt controllers. This arm, which is guided by machine learning classification output, sorts the collected items into dedicated bins. The disposal bins all feature ultrasonic sensors that constantly track and report fill levels. By embedding intelligence in this component of the system, the disposal is systematic and sustainable, reducing human interference and mistake in segregation of waste.

- **Mobile Application**: The mobile interface is the human-machine interface of the system. The disposal systems and the robot run independently, but the mobile app provides visibility and accountability. It displays actual fill levels for disposal bins, giving cleaning staff complete visibility of the system's status. Automatic notifications are issued when a bin is close to full capacity, prompting timely action. This ability closes the loop of autonomous and human monitoring, always remaining effective in protecting railway station sanitation.

Through the integration of the foregoing subsystems, the solution responds to several concerns simultaneously: autonomous garbage detection and pickup, free passage in crowded public areas, smart separation and sorting, and effective human-machine interaction. The combination of IoT, cloud-based intelligence, and robotics enables the railway stations to dynamically maintain cleanliness levels based on actual conditions rather than predetermined schedules. Coupling autonomy with intelligent notifications and real-time monitoring, the system offers a revolutionary approach to modern waste management of high-density public spaces.

3.2 Process Workflow

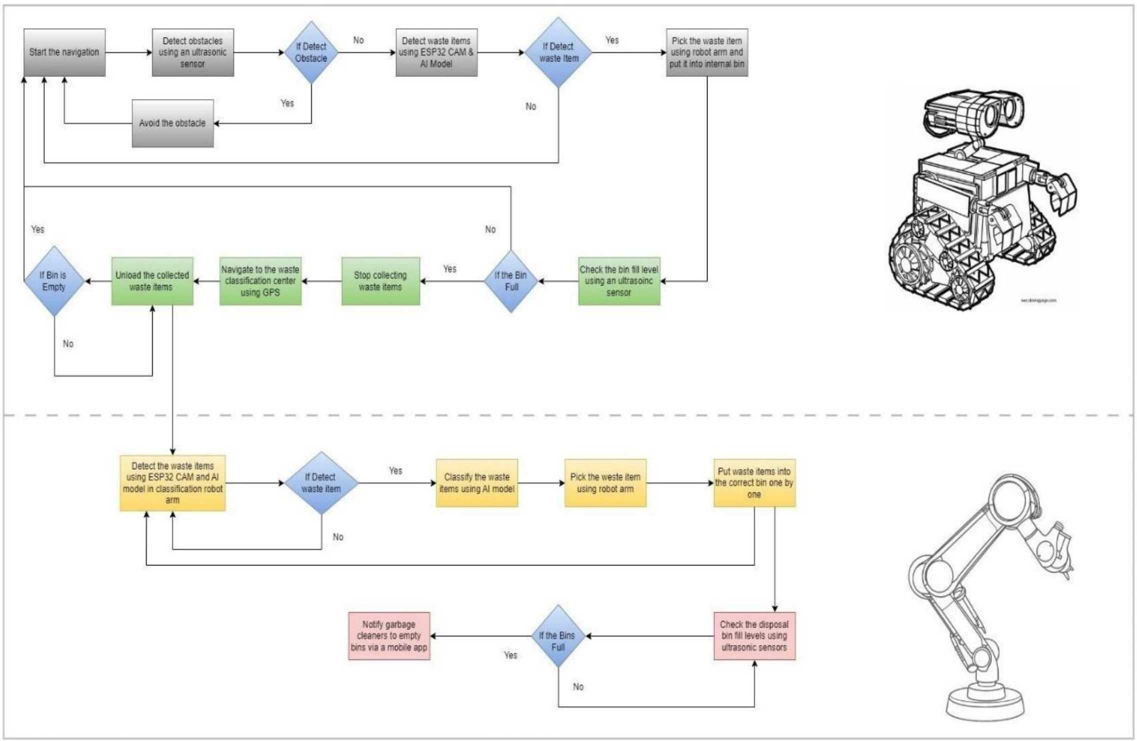

Figure 3.2: Flowchart for the overall system

The overall workflow of the proposed system has been designed to ensure that waste is identified, collected, transported, and classified autonomously and effectively with real-time monitoring and alert. The workflow begins with the mobile robot actively navigating around to identify waste objects. Using onboard sensors and an integrated vision model, the robot differentiates between potential garbage objects and irrelevant environmental elements. Once a valid garbage object is detected, the robotic arm is activated to grasp the object and place it in the robot's internal bin.

As the robot continues operation, the internal bin level is continuously monitored by ultrasonic sensing. When the bin is 80% (or a predetermined percentage) full, the system proceeds to the unloading phase. In this stage, the robot switches its task from active collection to navigation. The robot navigates to the disposal point via a combination of location awareness and obstacle avoidance so that the robot can safely operate within congested or semi-structured railway station environments.

Once the robot reaches the disposal area, it performs the unloading task, emptying collected waste onto a designated platform. The disposal subsystem is then activated, where the classification model separates each waste item into four categories Plastic, Paper, Glass, and Metal. The stationary robotic arm releases each item into its corresponding disposal container, which has been preconfigured and placed at precise locations for consistency.

Simultaneously, all the garbage bins are equipped with ultrasonic sensors to constantly monitor their levels. These are sent via the IoT layer to a cloud database. A mobile application makes this visual, providing cleaning staff with real-time feedback and notifications when a bin is reaching full capacity. This enables timely human intervention only when required, thereby avoiding spillage and unnecessary checking.

The *Figure 3.2* illustrates this end-to-end process, beginning with initial trash detection, followed by navigation and unloading, and concluding with final classification and notification. This structured workflow illustrates how multiple subsystems come together in unison to achieve the overall goal of autonomous waste management in railway stations.

3.3 Individual Component

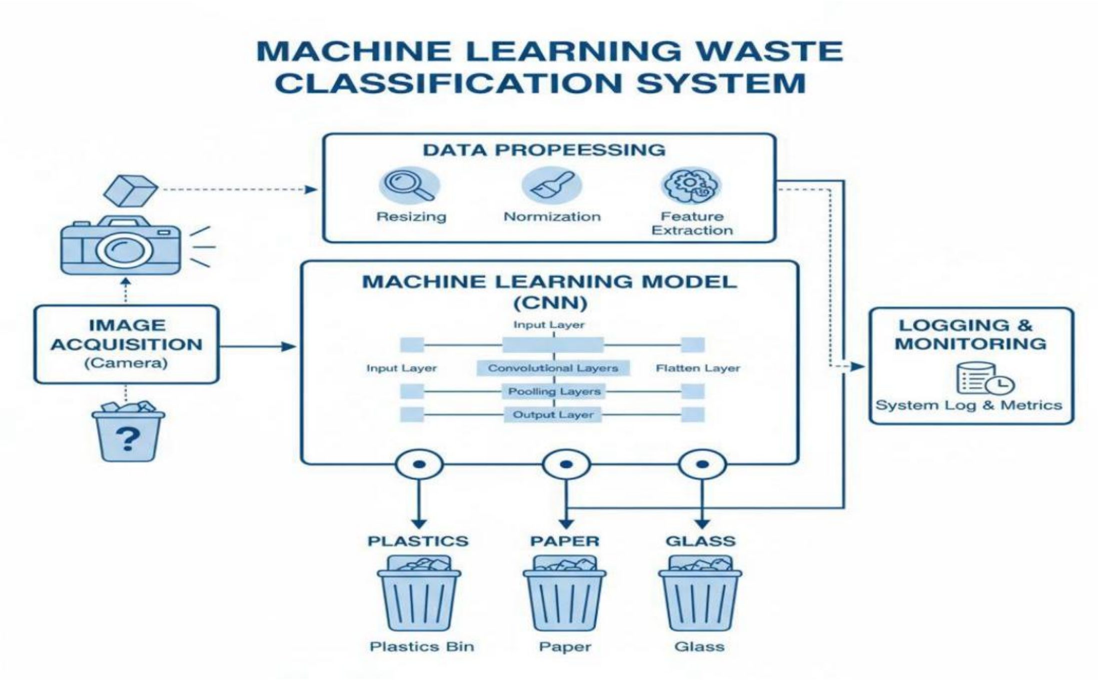

Figure 3.3: Individual component diagram

distance. The individual component of my research project is centered on the development of a machine learning-based waste classification system, designed to automatically categorize collected waste into four main categories: Plastic, Paper, Glass, and Metal. This component plays a vital role in the overall architecture of the garbage-collecting robot system, as it is responsible for ensuring that waste items collected from the railway platforms are accurately sorted before disposal. The waste classification process is implemented at the disposal station, which serves as the endpoint where the robot unloads the collected items. At this stage, the classification system takes over and operates independently to identify and categorize the waste for proper recycling or disposal.

The process begins with image capture, where a high-resolution camera mounted above the classification platform captures images of the waste items as they are unloaded. Controlled lighting is used to minimize the effects of environmental factors such as shadows, reflections, and low visibility, which could otherwise compromise the accuracy of image recognition. Once the raw images are collected, they undergo a preprocessing stage to enhance the quality and consistency of the input data. Preprocessing includes steps such as resizing images to a standard dimension, normalizing pixel values, and reducing noise. To further improve the model's robustness, data augmentation techniques are applied. These include rotating, flipping, and adjusting contrast levels in the images, which help the model generalize better when encountering waste items in different orientations or lighting conditions.

At the core of this component lies the machine learning model, specifically a Convolutional Neural Network (CNN), which is widely recognized for its superior performance in image classification tasks. The CNN is trained on a dataset containing labeled images of the four target waste categories: plastic, paper, glass, and metal. During training, the network learns to identify subtle visual features such as texture, edges, shapes, and color patterns that distinguish one type of waste from another. To improve the model's efficiency and accuracy, transfer learning techniques can be applied using pre-trained networks like ResNet, VGG, or MobileNet. These models, already trained on large-scale image datasets, provide a strong foundation by transferring generalized visual feature knowledge, which significantly reduces training time while boosting classification performance.

Once trained, the CNN performs classification and decision-making in real-time. For every input image, the model outputs a probability distribution across the four classes. The waste item is then assigned to the class with the highest probability score. For example, if the model predicts a 92% probability for "plastic," that item will be categorized as plastic waste. The classification result is then passed to the sorting mechanism, which physically directs the item into the appropriate disposal bin. This ensures that waste is not only collected but also systematically categorized for recycling or safe disposal.

In terms of system integration, the classification model functions as a modular component within the overall robotic system. It can communicate seamlessly with the robot and the disposal station through APIs or data exchange protocols. The classified results are logged into a local or cloud-based database, which serves multiple purposes such as system monitoring, performance evaluation, and dataset expansion. Over time, this data can be used to retrain the model, thereby improving its accuracy and adaptability to new waste types or environmental conditions.

The implementation of this component brings several significant advantages. It ensures consistency and efficiency in waste classification, eliminating the errors and inconsistencies often associated with manual sorting.

Furthermore, it reduces the need for human involvement in handling waste, thereby minimizing exposure to hazardous materials and improving occupational safety. The modularity of the system also makes it scalable and adaptable to various real-world environments, from railway platforms to urban waste management facilities. Finally, the system provides room for future expansion, as additional categories such as electronic waste or organic waste can be integrated into the model through retraining.

In conclusion, the machine learning-based waste classification component is not just a technical addition but a critical enabler of the entire robotic waste collection system. By leveraging advanced image recognition techniques, it ensures that waste is categorized accurately and efficiently, contributing to more sustainable and environmentally responsible waste management practices

3.4 Project Management and Development Approach

3.4.1 Requirements gathering and analysis

This phase focuses on understanding the project scope by identifying the functional and user requirements. Stakeholder discussions, literature review, and observation of existing waste management practices are used to define system needs and expectations..

3.4.2 Technology selection

In this step, the suitable tools, frameworks, and platforms for system development are identified. For the waste classification component, this includes selecting machine learning frameworks (TensorFlow), programming languages (Python), hardware (camera), and possible cloud platforms for model training.

3.4.3 Agile Development Process

The project adopts an Agile methodology, ensuring iterative development through sprints. Each sprint includes planning, implementation, testing, and review, allowing flexibility and gradual system improvement.

3.4.4 Task Allocation and Team Collaboration

Roles are assigned among team members to ensure efficiency. Each component is handled by responsible individuals, while supervisors provide continuous guidance and feedback..

3.5 Project Execution and Monitoring

3.5.1 Sprint Planning and Execution

Agile sprints are planned with clearly defined goals such as dataset preparation, model training, or integration testing. Each sprint typically spans two weeks, ensuring incremental progress.

3.5.2 Risk Assessment and Mitigation

Potential risks such as data set limitations, hardware failures, or model underperformance are identified early. Mitigation strategies include backup datasets, testing alternative models, and consulting supervisors.

3.5.3 Progress Tracking and Reporting

Progress is tracked using tools such as Gantt charts, sprint backlogs, or progress diaries. Weekly meetings with supervisors ensure the project stays aligned with objectives.

3.6 Testing and Validation

Testing is conducted at multiple stages to evaluate accuracy, reliability, and performance. This includes dataset validation, model testing with unseen data, and prototype evaluation in controlled environments.

3.6.1 Documentation and Review

Each development stage is documented for future reference and final reporting. Review sessions are conducted at the end of each sprint to refine the component and resolve issues before moving to the next stage.

3.7 Gantt Chart

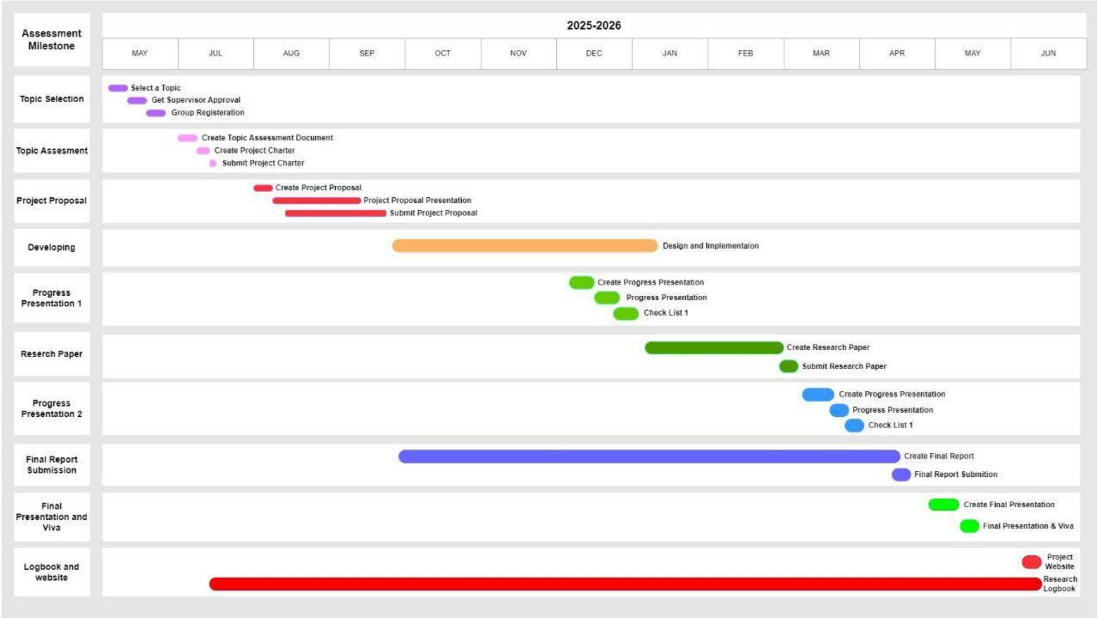

Figure 3.4: Gantt chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system shall preprocess the captured images (resizing, normalization, noise reduction).
- The system shall classify waste into four categories: plastic, paper, glass, and metal.
- The system shall display the classification results with confidence levels.
- The system shall direct the classified waste item to the correct bin.
- The system shall log classification outputs for evaluation and reporting.

4.2 Non-functional Requirements

- The system should achieve a classification accuracy of at least 85%.
- The system should provide near real-time classification results with minimal delay.
- The system should be reliable and available during operation without frequent failures.
- The system should be scalable to allow adding more waste categories in the future.
- The system should be user-friendly and require minimal technical expertise to operate.
- The system should operate within resource constraints (camera, computing power, cloud).

4.3 User Requirements

- The user should be able to easily operate the system without technical knowledge.

- The user should be able to monitor classification results clearly.

- The user should not require additional manual sorting once the system is running.

- The user should have confidence in the system’s accuracy and reliability.

- The user should be able to access logs or reports for verification of system output.

4.4 System Requirements

• A camera device must be available for image capture.

• A computer or microcontroller must be available for running the CNN model.

• Required software libraries (Python, TensorFlow/Keras, OpenCV, etc.) must be installed.

• Cloud services or a local GPU may be required for model training.

• The system must have storage capacity for datasets and logging outputs.

• The system should have proper integration with sorting mechanisms (bins, motors).

5. DESCRIPTION OF PERSONAL AND FACILITIES

This research project is carried out by Thisara Nuwanthi (IT22566102) under the guidance of the assigned supervisors. The work is supported with access to essential facilities provided by the university, including laboratory space for conducting experiments, supervision for academic and technical guidance, and access to necessary research materials. A personal computer and internet facilities are utilized for data collection, analysis, and documentation, while additional support such as camera equipment and cloud services may be used where required. These combined resources provide a suitable environment to successfully design, develop, and test the waste classification component of the research.

6. BUDGET AND BUDGET JUSTIFICATION

Table 6.1 Budget plan

Purpose	**Estimated Cost (LKR)**

Camera (To capture images of waste items)	11000
Cloud Services	5000
Total Estimated Cost	16000

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